

Research Paper 16 Summary

TROPOMI on the ESA Sentinel-5 Precursor: Chemical Signatures

Problem Statement: AI-Based Detection and Classification of Industrial Fires (PS26162)

Original Paper: Veeffkind et al. (2012), *Remote Sensing of Environment*

1 Basic Information

- **Paper Title:** TROPOMI on the ESA Sentinel-5 Precursor: A GMES Mission for Global Observations
- **Authors:** J. Pepijn Veeffkind, Ilse Aben, K. McMullan, et al. (2012)
- **Sensor Covered:** TROPOMI (TROPOspheric Monitoring Instrument) on the European Space Agency's Sentinel-5P satellite.

2 What is this Paper About? (In Simple Words)

While thermal satellites (like NASA FIRMS) can see where heat is, they cannot tell you what kind of smoke is rising into the air. This paper introduces TROPOMI, a satellite instrument that measures air pollution and atmospheric gases across the entire planet every single day.

- **How it Works:** TROPOMI uses high-resolution spectrometers (measuring Ultraviolet, Visible, Near-Infrared, and Shortwave Infrared light) to detect the unique chemical “fingerprints” of different gases in the air.
- **Key Gases Measured:**
 1. **Sulfur Dioxide (SO₂) & Nitrogen Dioxide (NO₂):** Produced in large volumes by industrial combustion (refineries, coal-fired power plants, metal smelters, and chemical complexes).
 2. **Carbon Monoxide (CO) & Smoke Aerosols:** Produced predominantly by wildland forest fires and agricultural stubble burning.

3 Key Concept: The Chemical Fingerprint Table

This paper enables our system to classify fires based on their chemical byproducts:

4 How This Helps Our Project (PS26162)

4.1 1. A Major Differentiator: Chemical Cross-Validation Layer

Most existing fire solutions look only at temperature. By ingesting daily Sentinel-5P TROPOMI data alongside NASA FIRMS, our system adds a multi-modal chemical validation layer:

Combustion Category	Dominant Chemical In Plume	Secondary Signals
Industrial Facilities / Refineries	High SO ₂ and High NO ₂	Low to Moderate CO
Thermal Power Plants / Smelters	Extreme SO ₂ Plumes	High NO ₂
Forest Fires / Wildfires	High CO & Carbon Aerosols	Near Zero SO ₂
Agricultural Stubble Burning	High CO & Organic Smoke	Near Zero SO ₂

- **Thermal Trigger (FIRMS):** Alerts our system to a hot spot at a specific coordinate.
- **Chemical Check (TROPOMI):** Checks the atmospheric column above that coordinate.
If an SO₂/NO₂ plume is present, it chemically confirms industrial combustion.

4.2 2. Flawless Separation of Stubble Burns from Refineries

During peak agricultural burning in India, thermal satellites get overwhelmed. Because crop stubble releases almost zero SO₂, whereas refineries and coal plants emit heavy SO₂, our chemical cross-check layer completely eliminates agricultural false positives.

4.3 3. Ingestion in GIS Map Overlays

Sentinel-5P data is freely accessible via the Copernicus Open Access Hub. Our GIS solution can overlay daily SO₂ and NO₂ heatmaps on top of the base map, providing visual proof to control-room operators.